OPTIMISATION

1. a) The function f is such that

$$f(x,y) = \left(x^2 - \frac{1}{2}\right)^2 + \left(y^2 - \frac{1}{2}\right)^2 - \frac{1}{2},$$

which shows that the function is radially unbounded. As a result, all its level sets are compact and the function has at least a global minimizer. Note that this is not unique and it is such that $f_{\min} = -1/2$. [2 marks]

b) The first order necessary conditions of optimality are

$$0 = \frac{\partial f}{\partial x} = 4x^3 - 2x, \qquad 0 = \frac{\partial f}{\partial y} = 4y^3 - 2y,$$

yielding the stationary points

$$P_1 : \left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right), \quad P_2 : \left(\frac{\sqrt{2}}{2}, 0\right), \quad P_3 : \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right),$$

$$P_4 : \left(0, -\frac{\sqrt{2}}{2}\right), \quad P_5 : (0,0), \quad P_6 : \left(0, \frac{\sqrt{2}}{2}\right),$$

$$P_7 : \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right), \quad P_8 : \left(-\frac{\sqrt{2}}{2}, 0\right), \quad P_9 : \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right),$$

[4 marks]

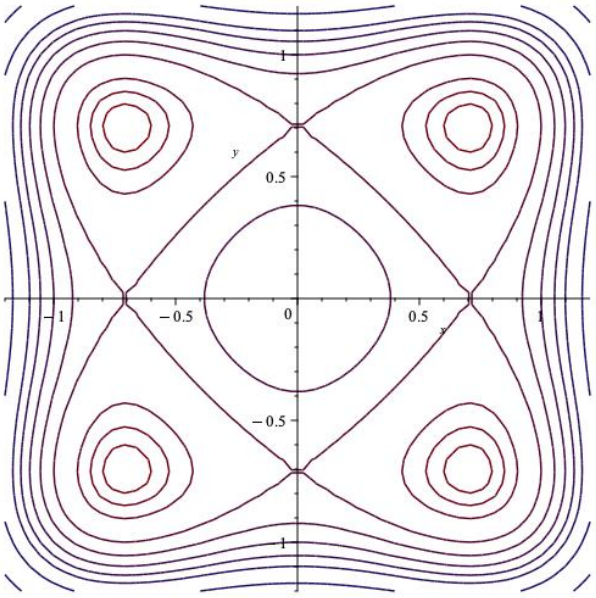
c) Note that

$$\nabla^2 f = \begin{bmatrix} 12x^2 - 2 & 0 \\ 0 & 12y^2 - 2 \end{bmatrix}.$$

Hence P_1, P_3, P_7, and P_9 are minimisers, P_5 is a maximiser, and P_2, P_4, P_6, and P_8 are saddle points. [4 marks]

d) Note that $f(P_1) = f(P_3) = f(P_7) = f(P_9) = -1/2$, hence all points are global minimiser. This is consistent with the result in part a). A sketch of the level lines is given in the plot below, which clearly show the location of the minimisers, the maximizer, and the saddle points and the fact that all level sets are compact.

Answers



[4 marks]

e) A direct calculation yields

$$x_{k+1} = 4\,\frac{x_k^3}{6x_k^2-1}, \qquad y_{k+1} = 4\,\frac{y_k^3}{6y_k^2-1},$$

hence $N(x) = 4\,\dfrac{x^3}{6x^2-1}$. [2 marks]

To check the speed of convergence of a sequence $\{x_k\}$ converging to zero consider the limit

$$\lim_{k\to\infty}\frac{\mathscr{E}_{k+1}}{\mathscr{E}_k^3} = \lim_{k\to\infty}\frac{\left|4\,\dfrac{x_k^3}{6x_k^2-1}\right|}{|x_k|^3} = 4,$$

yielding the indicated speed of convergence. [2 marks]

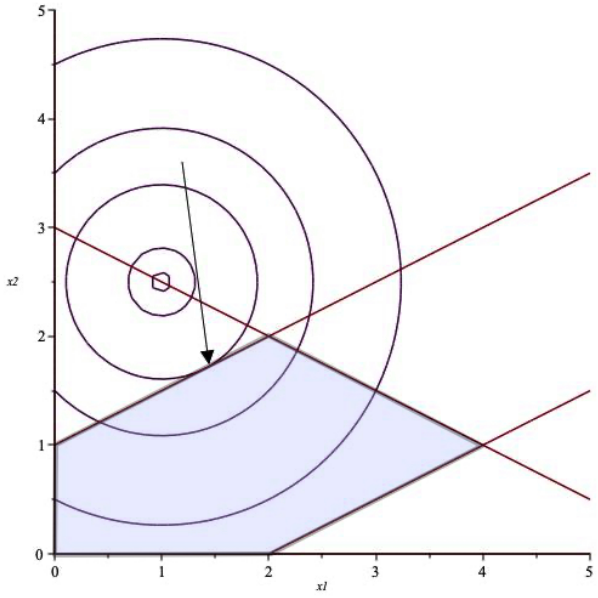
Similarly, to check the speed of convergence of a sequence $\{x_k\}$ converging to $\sqrt{2}/2$ consider the limit

$$\lim_{k\to\infty}\frac{\mathscr{E}_{k+1}}{\mathscr{E}_k^2} = \lim_{k\to\infty}\frac{\left|x_{k+1}-\dfrac{\sqrt{2}}{2}\right|}{\left|x_k-\dfrac{\sqrt{2}}{2}\right|^2} = 3\,\frac{\sqrt{2}}{2},$$

yielding the indicated speed of convergence. [2 marks]

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2. a) The level lines of the cost function and the admissible set are sketched in the graph below. Note that the admissible set is the closed shaded region.



[4 marks]

b) By definition all points in the interior of the admissible set are regular points. The points on the boundary of the admissible set and such that only one constraint is active are also regular points (the gradient of the active constraint is non-zero). Finally, the points on the boundary for which two constraints are active are regular since the constraints are linearly independent. [4 marks]

c) By inspection of the sketch of the level lines of the cost and of the admissible set one concludes that the optimal point (indicated by an arrow) is the point in which a level line of the cost is tangent to the line $x_1 - 2x_2 + 2 = 0$, which yields the optimal point $(x_1^\star, x_2^\star) = (7/5, 17/10)$. [4 marks]

d) The Lagrangian for this problem is

$$L(x_1, x_2, \rho_1, \rho_2, \rho_3, \rho_4, \rho_5) = (x_1 - 1)^2 + \left(x_2 - \frac{5}{2}\right)^2 + \rho_1(-x_1) + \rho_2(x_2) +$$

$$\rho_3(-x_1 + 2x_2 - 2) + \rho_4(x_1 + 2x_2 - 6) + \rho_5(x_1 - 2x_2 - 2).$$

Before writing the necessary conditions of optimality note that the optimal point is such that $-x_1^\star + 2x_2^\star - 2 = 0$, that is the third inequality constraint is active, whereas all other constraints are not active. As a result, the optimal solution is such that $\rho_1 = \rho_2 = \rho_4 = \rho_5 = 0$. The necessary conditions of optimality thus reduce to

$$0 = 2(x_1 - 1) + \rho_3(-1), \qquad 0 = 2\left(x_2 - \frac{5}{2}\right) + \rho_3(2), \qquad \rho_3 \geq 0.$$

Replacing the optimal point yields $\rho_3 = 4/5$. [4 marks]

e) As already noted, only the third constraint is active and for this $\rho_3 > 0$. All other inequality constraints are not active. As a result, the strict complementarity conditions hold. This allows applying the second order sufficient conditions of optimality. Note, however, that $\nabla^2 \left[(x_1 - 1)^2 + \left(x_2 - \frac{5}{2}\right)^2\right] = 2I > 0$, hence

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(without computing the Jacobian of the active constraint) one concludes that the point identified in part c) is a local minimiser for the considered problem.

[4 marks]

Answers

3. a) The Lagrangian of the problem is

$$L(x,y,z,\lambda_1,\lambda_2) = (x+1)^2 + (y+1)^2 + z^2 + \lambda_1 x + \lambda_2 y,$$

yielding the necessary conditions of optimality

$$0 = 2(x+1) + \lambda_1, \quad 0 = 2(y+1) + \lambda_2 \quad 0 = 2z, \quad 0 = x, \quad 0 = y.$$

The only candidate point is therefore $x = 0$, $y = 0$, $z = 0$ with multipliers $\lambda_1 = -2$ and $\lambda_2 = -2$. [4 marks]

b) The Lagrangian of the problem is

$$L(x,y,z,\lambda_1,\lambda_2) = (x+1)^2 + (y+1)^2 + z^2 + \lambda_1 x^2 + \lambda_2 y^2,$$

yielding the necessary conditions of optimality

$$0 = 2(x+1) + 2\lambda_1 x, \quad 0 = 2(y+1) + 2\lambda_2 y \quad 0 = 2z, \quad 0 = x^2, \quad 0 = y^2.$$

These conditions do not give any candidate solution. This is due to the fact that all points of the admissible set, that is the points on the z-axis, are not regular points, since the gradients of the active constraints are both zero there.
[4 marks]

c) The sequential penalty function for the problem P is

$$F_\varepsilon(x,y,z) = (x+1)^2 + (y+1)^2 + z^2 + \frac{1}{\varepsilon}(x^2+y^2).$$

This is a quadratic function with a global minimiser, for all $\varepsilon > 0$, given by

$$p_\varepsilon = \left(-\frac{\varepsilon}{1+\varepsilon}, -\frac{\varepsilon}{1+\varepsilon}, 0\right).$$

As $\varepsilon \to 0$ the point p_ε converges to the candidate optimal solution determined in part a). The corresponding approximate multipliers are

$$\lambda_{1,\varepsilon} = -\frac{2}{\varepsilon+1}, \qquad \lambda_{2,\varepsilon} = -\frac{2}{\varepsilon+1},$$

which converge, as $\varepsilon \to 0$, to the optimal multipliers. [4 marks]

d) The exact penalty function for the problem P is

$$G(x,y,z) = (x+1)^2 + (y+1)^2 + z^2 - 2x(x+1) - 2y(y+1) + \frac{1}{\varepsilon}(x^2+y^2).$$

This is a quadratic function with a unique stationary point (for $\varepsilon \neq 1$), that is the point $x = 0$, $y = 0$, $z = 0$, which coincides with the optimal solution of problem P. Note, finally, that

$$\nabla^2 G = \operatorname{diag}\left(\frac{2}{\varepsilon} - 2,\ \frac{2}{\varepsilon} - 2,\ 2\right),$$

hence the stationary point is a minimiser provided $\varepsilon < 1$.

Note that for $\varepsilon = 1$ the function G has infinitely many stationary points, all of which are (non-strict) minimisers. [4 marks]

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e) The sequential penalty function for the problem $\tilde{P}$ is

$$\tilde{F}_\varepsilon(x,y,z) = (x+1)^2 + (y+1)^2 + z^2 + \frac{1}{\varepsilon}(x^4 + y^4).$$

The Hessian of this function is (strictly) positive definite for all (x,y,z) and all $\varepsilon > 0$, hence the function is (strictly) convex. (The function $\tilde{F}_\varepsilon$ is also the sum of convex functions, hence it is convex.) The function is, in addition, radially unbounded, hence bounded from below. As a result the function has a global minimizer, which is also a stationary point. Such a global minimizer is such that $\nabla \tilde{F}_\varepsilon = 0$, that is

$$0 = 2(x+1) + 4\,\frac{x^3}{\varepsilon}, \quad 0 = 2(y+1) + 4\,\frac{y^3}{\varepsilon}, \quad 0 = z.$$

As $\varepsilon \to 0$, the only solution of these equations converges to $(x,y,z) = (0,0,0)$ (as an $O(\varepsilon^{1/3})$), hence to the solution of problem $\tilde{P}$, which is the same as the solution of problem P. [4 marks]

Answers

4. a) The Lagrangian of the problem is

$$L(x_1,x_2,x_3,\lambda,\rho_1,\rho_2,\rho_3) = c_1(x_1)+c_2(x_2)+c_3(x_3)+$$
$$\lambda(x_1+x_2+x_3-r)+\rho_1(-x_1)+\rho_2(-x_2)+\rho_3(-x_3).$$

i) The first order necessary conditions of optimality are

$$0=2(x_1+1)+\lambda-\rho_1, \quad 0=4(x_2+1)+\lambda-\rho_2, \quad 0=6(x_3+1)+\lambda-\rho_3,$$
$$0=x_1+x_2+x_3-r$$
$$-x_1\le 0, \quad -x_2\le 0, \quad -x_3\le 0, \quad \rho_1\ge 0, \quad \rho_2\ge 0, \quad \rho_3\ge 0,$$
$$\rho_1x_1=0, \quad \rho_2x_2=0, \quad \rho_3x_3=0.$$

[4 marks]

ii) Optimality of the indicated point requires $\rho_1=\rho_2=0$. This yields

$$x_1=\frac{1+2r}{3}, \quad x_2=\frac{r-1}{3}, \quad \lambda=-\frac{8+4r}{3}, \quad \rho_3=\frac{10-4r}{3}.$$

As a result, any $r\in(1,5/2]$ yields a candidate optimal solution of the required form.

b) The Lagrangian for the problem is

$$L(x,y,\lambda_1,\lambda_2)=1+\lambda_1(x^3-3xy^2+1)+\lambda_2(3x^2y-y^3).$$

i) The necessary conditions of optimality are

$$0=3(x^2-y^2)\lambda_1+6xy\lambda_2, \quad 0=-6xy\lambda_1+3(x^2-y^2)\lambda_2$$
$$0=x^3-3xy^2+1, \qquad 0=3x^2y-y^3.$$

[4 marks]

ii) Using the last equation yields two cases.

C1: $y=0$. This yields $x=-1$ and $\lambda_1=\lambda_2=0$.

C2: $3x^2-y^2=0$. This yields $x=1/2$, $y=\pm\frac{\sqrt{3}}{2}$, and $\lambda_1=\lambda_2=0$.

In summary, we have three candidate solutions which give the cubic roots of -1, as requested. [4 marks]

iii) Modifying the cost from $f=1$ to an arbitrary function does not modify the problem. In fact, the admissible set of problem Z is composed of three isolated points, that is the cubic roots of -1, hence these are candidate optimal solutions for any f. Note, however, that the optimal multipliers depend upon the selection of f. [4 marks]

Answers